

Simple symmetric Venn diagrams with 17, 19 and 23 curves

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Abstract

We exhibit simple, rotationally symmetric Venn diagrams with 17 curves, with 19 curves and with 23 curves: n Jordan curves carried to one another by rotation through $2\pi/n$, with every one of the 2^n regions present and connected and, since the diagrams are simple, every crossing on exactly two curves. Symmetric Venn diagrams exist for every prime number of curves (Griggs, Killian and Savage, 2004), but those diagrams have many curves through a point; simple ones were known only up to 13 curves (Mamakani and Ruskey, 2014). Four 17-curve, nine 19-curve and five 23-curve diagrams were found by a Metropolis walk on rotation-invariant quadrangulations of the sphere in which regions may temporarily be duplicated, started from the Griggs–Killian–Savage diagram with its multiple crossings resolved. At 23 curves the walk was held for two weeks by duplicated regions near the poles; the two lineages that finished were the first whose $E = 92$ states had none. Every diagram is given by a machine-checkable certificate; one certificate each of the 17- and 19-curve sizes has been verified by a formal proof in Lean 4. All of the diagrams are non-monotone, which is why the crossing-sequence searches that found the 11- and 13-curve diagrams could not have found them.

1 Statement

An n -Venn diagram is a family of n Jordan curves in the plane such that for every subset S of the curves the set of points inside exactly the curves of S is nonempty and connected. It is *simple* if no point lies on three curves, and *symmetric* if a rotation by $2\pi/n$ maps the family to itself. Henderson observed in 1963 that a symmetric n -Venn diagram can exist only for prime n [1]; his argument had a gap, and the first complete proof of his theorem is due to Wagon and Webb [2]. Grünbaum asked in 1975 whether they exist for every prime and gave simple symmetric diagrams with five curves [3]; simple symmetric diagrams with seven curves were found by Grünbaum and by Edwards in the 1990s [4, 5]. Hamburger constructed the first symmetric 11-Venn diagram in 2002 [7]; it is not simple, and the same holds for the diagrams of Griggs, Killian and Savage, who then proved that symmetric n -Venn diagrams exist for every prime n [9] (see also [10]). Simple symmetric diagrams with 11 and 13 curves were found by Mamakani and Ruskey in 2012 and 2014 [13, 15] by a search over monotone diagrams, and the question for larger primes, put forward as the central open problem by Ruskey, Savage and Wagon [12], remained open: the recent literature lists simple symmetric n -Venn diagrams as known only for $n \leq 13$ [16]. The survey of Ruskey and Weston [11] has the full history.

Theorem 1. *Simple symmetric Venn diagrams with 17 curves exist, and so do simple symmetric Venn diagrams with 19 curves and with 23 curves.*

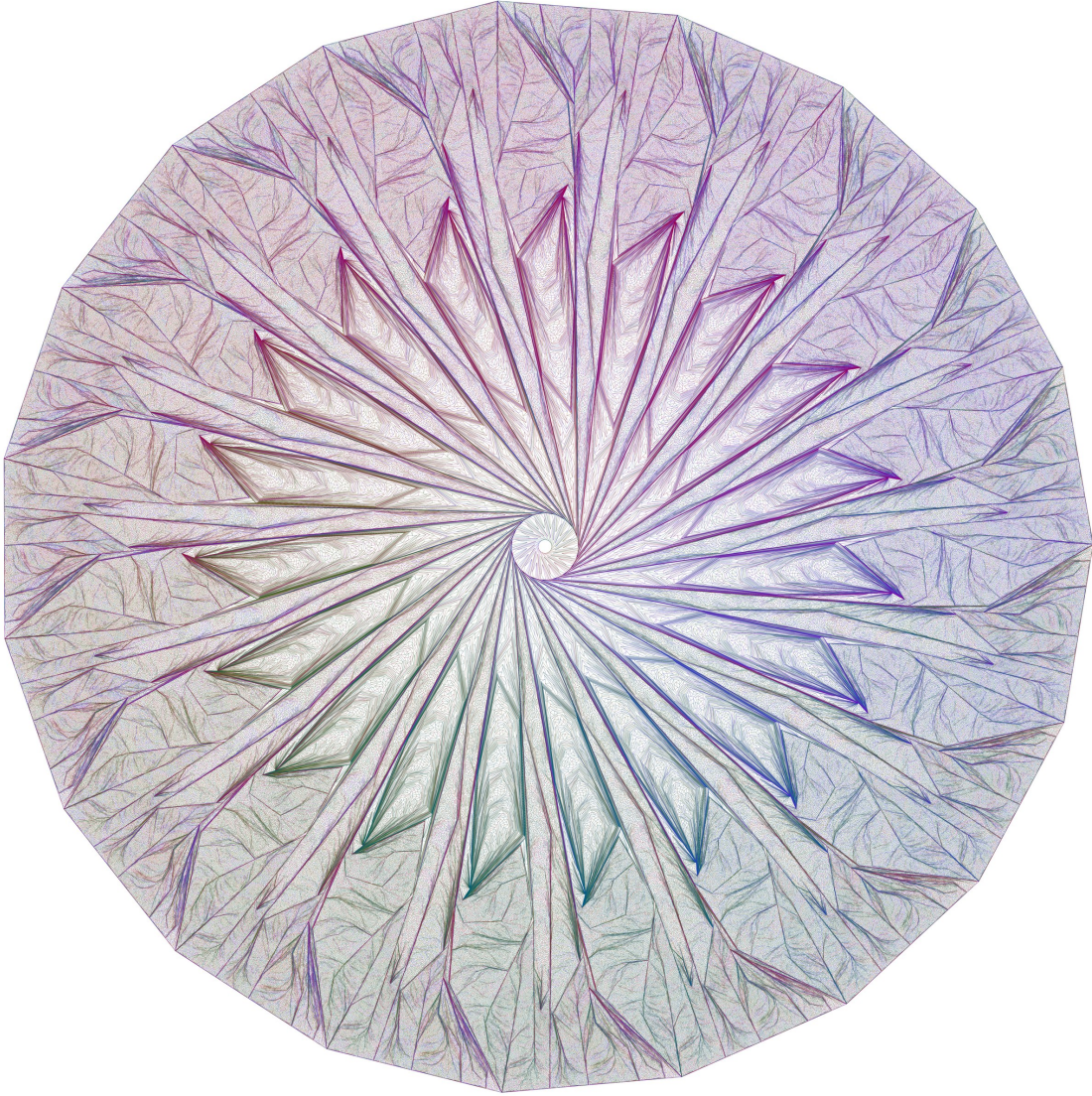


Figure 1: One of the nine 19-curve diagrams (certificate `fc7ad34`), drawn from its certificate at near-uniform crossing density by the plotter described in the data repository: 19 curves, 524,286 crossings, every one of the 524,288 regions present exactly once. The centre hole is the region inside all 19 curves; the outer face is the region outside all of them. Colours identify curves.

The proof is by exhibition. Four 17-curve, nine 19-curve and five 23-curve diagrams are given as certificates in the form of Section 2, each accepted by two independently written checkers, and one each of the 17- and 19-curve certificates has been formally verified in Lean 4 (Section 3). The 17- and 19-curve diagrams were found on 17 and 20–22 September 2026 and reported in the first version of this note; the 23-curve diagrams were found on 5 and 6 October 2026 and are described in Section 5.

2 Encoding and verification

Let D be a simple symmetric n -Venn diagram, drawn on the sphere with the unbounded region as a face. Its dual map M has one vertex per region, labelled by the region’s membership pattern in $\{0, 1\}^n$; one edge per arc, joining labels that differ in exactly the bit of the arc’s curve; and one face per crossing, a 4-cycle with labels $u, u \oplus e_a, u \oplus e_a \oplus e_b, u \oplus e_b$ (a *cube square*). Conversely:

Proposition 1. *A map on the sphere whose vertices are labelled by distinct elements of $\{0, 1\}^n$, all 2^n occurring, whose faces are cube squares, whose edges of each coordinate i form a single cycle through the faces containing them, and whose vertex rotations are single cycles, is the dual of a simple n -Venn diagram; if the label rotation $x \mapsto (x_1, \dots, x_{n-1}, x_0)$ is an orientation-preserving automorphism of the map, the diagram is symmetric.*

Proof sketch. The map is a cellular embedding on the sphere; its geometric dual is a 4-regular plane graph whose vertices are the faces (crossings). The edges of coordinate i , chained through opposite sides of each crossing, form one closed curve C_i ; regions of the arrangement are the vertices of M , and by the label condition each point of $\{0, 1\}^n$ is realized by exactly one region, which is a disk. Rotation of the labels acts as a rotation of the sphere fixing the two vertices 0^n and 1^n ; taking 0^n as the outer face gives a plane diagram with n -fold rotational symmetry. $\square$

A certificate is the list of the $2^n - 2$ faces of M , each as four n -bit labels: 131,070 faces for $n = 17$, 524,286 for $n = 19$ and 8,388,606 for $n = 23$ (889 MB as JSON). Verification recomputes: all 2^n labels present exactly once, every edge in exactly two faces ($2^{n+1} - 4$ edges), Euler characteristic 2, every face transversal, a single rotation cycle at every vertex, one cycle per curve with both sides connected, and invariance under label rotation. This takes seconds for 17 and 19 curves; on the five 23-curve certificates the checker in the data repository took 693, 734, 610, 633 and 614 seconds with peak memory of 20.0, 21.4, 21.4, 22.5 and 21.4 GB. Two independently written checkers, a half-edge implementation in C++ and a face-set implementation in Python, accept all eighteen certificates. A third, written later and sharing no code with either, audits the raw search state directly; it has been run on the nine 19-curve certificates and on one of the four 17-curve certificates (`venn17-gcp-s16`, `431857aa`), whose raw state it reports as a valid instance embedding with 17 curves, 131,072 vertex instances and Euler characteristic 2. On the five 23-curve raw states, Codex’s native inspector, a program pinned by its hash that reimports the raw state, rebuilds the full graph and checks the vertex, edge and face counts, the Euler characteristic and the rotation orbits, with the input’s SHA-256 checked before and after each run, reported order 23, energy 0 and no missing or duplicated label on 6 October (c25 at 06:16, c19, c21 and the pilot diagram at 06:17–06:19, f4 at 08:15; the c25 receipt was reassessed after an error in the wrapper’s orbit formula, which had assumed that every vertex orbit has 23 members, with no native rerun).

3 Formal verification

The first 17-curve certificate (c178d7bd) was formally verified in Lean 4 with Mathlib by Justin Grimes [17], working from the definitions alone and without access to our code. His theorem states that the 17 curves constructed from the certificate are Jordan curves in the Euclidean plane, carried to one another by a rigid rotation through $2\pi/17$, with every one of the 2^{17} regions path-connected, no point on three curves, and every crossing a transversal crossing of two curves. The finite parts of the proof use Lean’s compiled evaluation (`native_decide`); the development builds from scratch in under an hour.

The first 19-curve certificate (fcb7ad34; JSON form ed26b3ba) was verified by a port of the same development to $n = 19$, carried out by Codex (OpenAI) from Grimes’s sources: the dimension-dependent constants were changed, the finite meridian and sector certificates that the proof consumes were regenerated for 19, and the vendored topology library was left untouched. The port builds in twenty minutes and proves `exists_simple_rotational_venn_19`: there exist $C : \text{Fin } 19 \rightarrow \mathcal{P}(\mathbb{R}^2)$ and sides S with `SimpleRotationalVenn C S`, the same predicate as at 17. Its axiom list is that of the 17-curve proof under the namespace rename: `propext`, `Classical.choice`, `Quot.sound` and the twenty `native_decide` certificates. We re-ran `#print axioms` on the built project to confirm this.

The same port has been prepared for $n = 23$ and takes the search engine’s certificate export directly; its default memory and time limits (8 GiB and one hour per stage) are not sized for 23 and must be raised. A run on the c25 certificate started on 6 October passed the port’s compiled finite checker (all of the finite combinatorial checks: 8,388,608 regions, 16,777,212 arcs, 8,388,606 crossings) and generated the meridian and sector witnesses that the proof consumes, but the full proof build stopped at that run’s 16 GiB memory cap; a resumed build from the same cache and witnesses, with a 64 GiB limit and no change to the statements or axioms, had been started at 08:17 on 6 October and was still running when this version was prepared. No formal proof of a 23-curve certificate had completed when this version was prepared.

4 Method

We search the space of rotation-invariant cube-square quadrangulations of the sphere in which several vertices may carry the same label. The energy of a state is $E = (\text{missing labels}) + \lambda (\text{duplicate regions})$; $E = 0$ with $\lambda > 0$ is exactly a certificate. Moves are the local Reidemeister-type rewirings of the dual (lens insertion and removal on a 3-cube, triangle flip, bigon insertion and removal), each applied simultaneously to all n rotation images; every move preserves the sphere, the cube-square faces and the one-cycle-per-curve property, so the walk never leaves the space of valid symmetric arrangements. Proposals are accepted by the Metropolis rule at temperature T , with half of the proposals targeted at a missing label or a duplicated region.

Allowing duplicates is what makes the search work: with duplicates forbidden, greedy growth of the same quadrangulations reaches 98.3% of the labels and jams, because the last lens insertions require two missing labels to coincide geometrically. With duplicates allowed, a missing label can be created next to a duplicate at zero cost and defects diffuse until they annihilate.

17 curves. The walk starts from the Griggs–Killian–Savage diagram [9] for $n = 17$ with every multiple crossing resolved into simple crossings, a state with all labels present and 69,394 duplicate regions. With λ ramped from 0.3 to 1 over three hours at $T = 0.25$, three of fifteen

independent runs reached $E = 0$; a fourth diagram was obtained by a further 40-minute cycle from a state with 68 duplicates. Ten later instrumented runs of the same recipe finished seven times within three hours, first reaching $E = 2n$ after $1.6\text{--}2.8 \times 10^9$ proposals and finishing within 1.8×10^9 more. The same method finds 11- and 13-curve diagrams from the trivial starting state in seconds and minutes: on a laptop busy with other work, 200 of 200 seeds produced a simple symmetric 11-Venn diagram, in a median of 0.8 seconds and at most 16 seconds, all 200 pairwise non-isomorphic (up to cyclic relabelling, mirror image and swapping the poles) and all non-monotone; 11 of 13 seeds produced a 13-curve diagram within a ten-minute limit, in a median of 52 seconds, again all distinct and all non-monotone.

19 curves: budget. The resolved starting diagram for $n = 19$ has all labels and 305,862 duplicate regions. Scaled by the number of regions, no 19-curve run in the first three days received the budget that the 17-curve runs needed: the longest had 6×10^9 proposals and was still descending when stopped, and every restart from a low-energy state was shorter than the 17-curve endgame itself.

19 curves: the knot. Every lineage that reached low energy stalled with the same duplicated label, $\{0, 3, 4\}$, whose two instances had degrees 6–9 and shared a neighbour. A region of the dual can move by a triangle flip only at degree 3, so the walk’s targeted proposals, when they landed on such a duplicate, did nothing. The duplicate is inherited: in the resolved starting diagram the label $\{0, 3, 4\}$ already appears twice, once as a region of degree 18. (The resolved 17-curve diagram has the same feature for $\{0, 1, 4\}$, degrees 4 and 17, and the 17-curve walk happens to grind it down; the ten 17-curve pre-completion states all have their duplicates at degree 3 or 4.) One added proposal class removes the obstruction: when a targeted proposal lands on a duplicate of degree above 3, it attacks that degree, by a triangle flip at a degree-3 neighbour or a lens insertion on an incident edge, both ordinary moves under the ordinary acceptance rule. From a frozen $E = 38$ state, 25 minutes at $\lambda = 0.35$ with this proposal reduced $\{0, 3, 4\}$ to multiplicity one; without it, 45 minutes at $\lambda = 0.30$ left it untouched. A three-hour ramp $\lambda : 0.5 \rightarrow 1$ then cooled the state to $E = 38$ with two missing labels of rank 11 and no duplicates.

19 curves: closure. From that state, continuations differing only in the random seed were run with λ held at 1 and with λ ramped from 0.6, 0.8 or 0.9 to 1. Held at 1, five of eight continuations reached $E = 0$, after 4 seconds, 13 seconds, 190 seconds, 4.3 hours and 6.5 hours; of the sixteen ramped continuations one finished, at the end of its six-hour ramp. The pre-completion states are saved exactly, by journal undo of the winning move inside the engine, and show three closing mechanisms: a single triangle flip of the last duplicated region, of degree 3, into the last missing label at Hamming distance exactly 3 (four diagrams, and one of the fresh-start diagrams below); a single lens insertion creating both missing labels at once after they had drifted to Hamming distance 1 (one diagram, and two of the fresh-start diagrams below); and a single lens removal of two adjacent duplicated regions in a state with no missing label (one diagram).

19 curves: fresh start. The six diagrams above descend from one hand-off state. Three more do not. Thirty-two 36-hour runs were started from the resolved scaffold on 20 September: twenty with the ordinary proposals (twelve on the fleet and eight on a laptop), six with the degree-attacking proposal applied to three in ten of the targeted landings at $T = 0.25$, and six

with it applied always at $T = 0.28$, all with λ ramped linearly from 0.3 to 1 over the wall time. Three reached $E = 0$: two of the three-in-ten runs, after 32.6 and 33.3 hours, and one laptop run with the ordinary proposals, after 34.0 hours, at 1.1×10^{10} to 2.1×10^{10} proposals. The first closed by the triangle flip above, from a state with one missing label of rank 8 and one duplicated label of rank 11, both copies of degree 3, at Hamming distance 3; the other two closed by the lens insertion, from states with no duplicate and two missing labels at Hamming distance 1. Each of the three shares only about a tenth of its faces with any other diagram in the table. When the September version of this note was written the remaining three-in-ten runs stood at best energies 38, 38, 114 and 152, the ordinary runs between 38 and 228, and the always-on runs no lower than 114.

5 23 curves

Times in this section are Eastern Daylight Time (UTC−4); the dated run records (the launch manifest and the cloud log), which are kept by the author, carry UTC as well.

Scale and engine. The resolved Griggs–Killian–Savage diagram for $n = 23$ has all 8,388,608 labels and 5,323,948 duplicate regions, seventeen times the 19-curve count, and a full state occupies 1.4 GB on disk. A second starting state, the *orbit-safe* scaffold, was prepared by Codex from the same diagram: of the 54 duplicated labels of low or high rank that had an instance of degree 8 or more (the 23-curve counterparts of the inherited $\{0, 3, 4\}$ at 19, which is one of them), 33 were removed before the walk began, the total number of duplicate regions being unchanged. The walk itself was rewritten by Codex to store one representative of each rotation orbit and apply every move through the rotation, with the same moves, energy, acceptance rule and export format; on a fixed workload it ran 3.3 times faster than the full-state engine at 23 curves in 1.1 GB of memory. A control at 17 curves on 29 September, eight seeds of the original engine against eight of the rewrite on the three-hour recipe of Section 4, closed 5 of 8 and 8 of 8 within the wall, so the difficulties below are not an artefact of the rewrite. In the walk’s own units the energy at $\lambda = 1$ is always a multiple of $2n = 46$, as it was a multiple of 38 at 19, and $E = 46$ is the last rung before a certificate.

The crust. The 23-curve campaign ran from 20 September to 6 October on one laptop and a succession of cloud instances. For two weeks every lineage that reached low energy stopped at $E = 230, 138, 92$ or 46 and stayed. A census of the duplicated labels in every stalled state we examined showed the same pattern: at least one duplicated label in the *crust*, rank at most 4 or at least $n - 4$. The clearest case is the lineage c5, an eight-day λ ramp from the orbit-safe scaffold followed by a 24-hour cooling from $T = 0.25$ to 0.14: it reached $E = 46$ on 1 October with the label $\{0, 7, 9\}$ duplicated at degrees 4 and 4, where the triangle flip cannot reach it, and eleven continuations on the cloud held that state for three days without improvement; a melting test (twelve chains, the degree-attacking proposal at lower λ for three different durations, then a ramp back) left the label at two copies in 28 of 28 saved states. Its sibling c8, from the standard scaffold, sat at $E = 92$ with $\{0, 3, 4\}$ and three labels of rank 20–21 duplicated, and 21 continuations for three days never reached $E = 46$. Twenty-four fresh coolings from c5’s end-of-ramp state, where $\{0, 7, 9\}$ was still single, avoided that knot in nine of the thirteen that reached $E = 46$, but those nine carried crust pairs instead ($\{0, 2, 5, 6\}$ with $\{0, 8\}$, or $\{0, 3, 4\}$ with $\{0, 2, 5, 6\}$), and none went below 46 in the 6–13 hours before that instance’s budget ended.

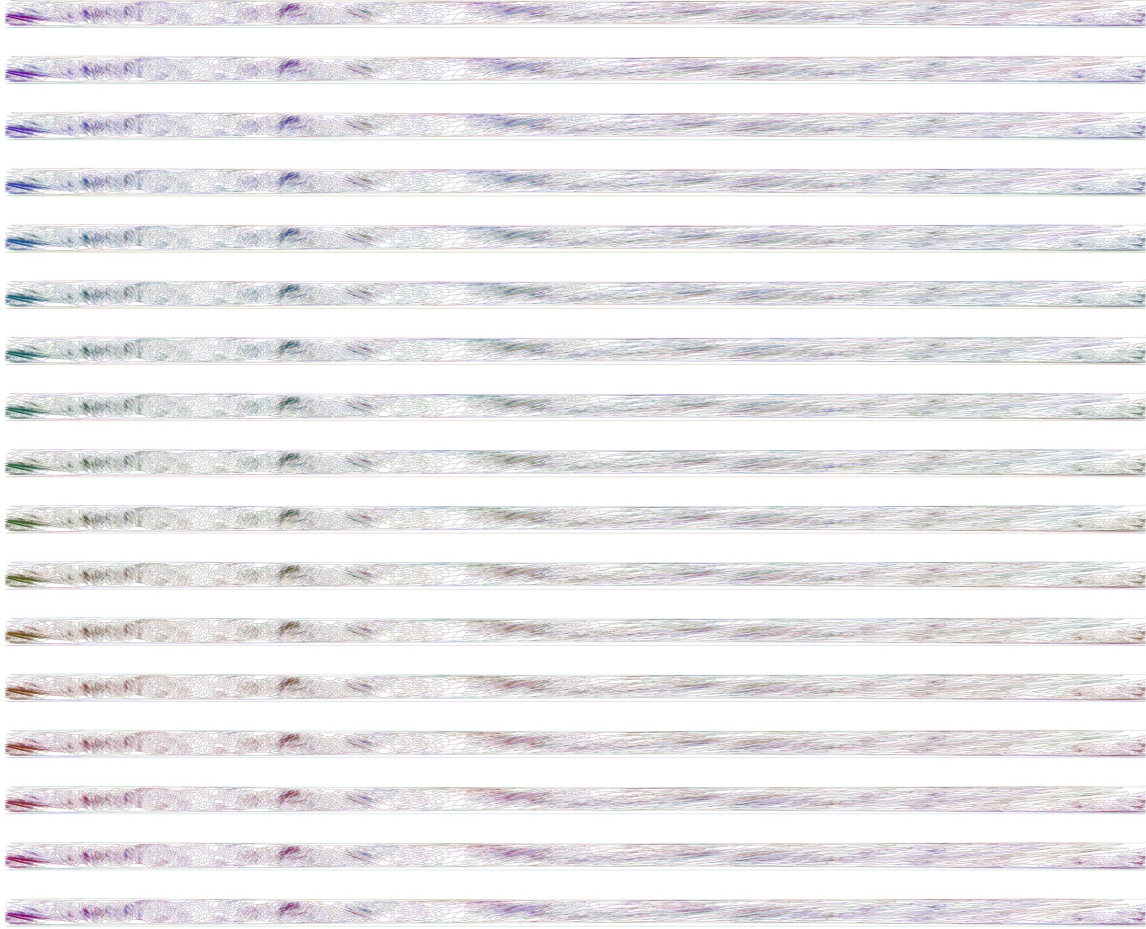


Figure 2: The 17-curve diagram `c178d7bd` unrolled: its seventeen congruent sectors, each a fundamental domain of the rotation containing $131,070/17 = 7,710$ crossings, drawn as rows, with the colour of each curve advancing by one position from row to row. The whole diagram is one row bent around the centre seventeen times.

The two lineages that did finish, c6 (orbit-safe scaffold) and c9 (standard scaffold), had the longest ramps of any lineage that reached low energy, and were the first whose $E = 92$ states had no crust duplicate at all.

The recipe. Both lineages were launched on 22 September with the degree-attacking proposal applied to three in ten of the targeted landings and λ ramped linearly from 0.3 at $T = 0.25$; the ramp was re-based that evening, about twenty hours after launch, to reach $\lambda = 1$ after twelve days, which it did on 4 October at engine time 1,101,666 seconds (c6) and 1,101,665 seconds (c9), with $E = 3,082$ (c6) and 3,036 (c9). After about an hour’s hold at $\lambda = 1$ and $T = 0.25$ (3,499 seconds for c6, 3,373 for c9), a 24-hour cooling from $T = 0.25$ to 0.14 at $\lambda = 1$ followed, from 22:21 on 4 October. Sixteen to seventeen hours into the cooling, on 5 October, both crossed $E = 92$ (c9 at 14:16, c6 at 15:07) with two duplicated labels of rank 8–12 and two missing labels of rank 9–16, and no crust defect. Those states were saved, and continuations were started from them, and from the $E = 46$ states they soon produced, with T held at 0.14 and λ at 1: six on the laptop (three at 19:02, two at 20:45, one at 21:12) and three on a cloud instance; three more (two on the laptop, one on the cloud) ran the 19-curve closure recipe, $T = 0.25$ with the degree-attacking proposal applied always. Four of these finished, three on the laptop and one on the cloud, and the pilot diagram, described below, came from a run that was not among these continuations. c25, from c6’s $E = 92$ state, reached $E = 46$ after 81 minutes (two duplicated labels, no missing), passed through the form with one duplicate and one missing label, rose to $E = 92$ again, and reached $E = 0$ at engine time 9,985 seconds, at 00:00:09 on 6 October, 2 h 47 min 41 s after launch. c19, from a state of c9 with one missing and three duplicated labels, spent six hours between $E = 92$ and 184; its $E = 46$ state was exported at 01:16, about 22,400 seconds after launch (the log’s progress lines are 1,800 seconds apart and first report $E = 46$ at engine time 23,413 seconds), and it reached $E = 0$ at engine time 25,892 seconds, at 02:15 on 6 October, 7 h 12 min 33 s after launch. c21 descends from c6 by a different route: c20, launched at 19:02 from c6’s $E = 92$ state, reached $E = 46$ sixteen minutes later in the form with one duplicated and one missing label, and c21 was launched at 20:45 from that state. Its log shows it moving between the two forms of $E = 46$ (one duplicate and one missing label; two duplicated labels and none missing) with excursions to $E = 92$, and it reached $E = 0$ at engine time 29,528 seconds, at 04:58 on 6 October, 8 h 13 min 18 s after launch.

The fourth diagram to be verified was found before any of these, by a run that was not among the continuations above and whose result was read only after the three above had been found and verified; we call it the pilot diagram. On the evening of 5 October a paired pilot of Codex’s memory-assisted engine (see “What did not work” below) was launched on the laptop from an earlier state of c6, the $E = 184$ export of its cooling (SHA-256 prefix 87b75d5b): two arms in sequence, first an ordinary Metropolis walk with the engine’s memory features inactive (its run record shows no tabu rejection, no backtrack and no aspiration commit), then a backtracking arm, each with a wall of 11,400 seconds, at $T = 0.14$ and $\lambda = 1$, seed 1065101, launched at 21:37. The ordinary arm reached $E = 46$ after 2,065 search seconds, in the form with two missing labels and no duplicate, and $E = 0$ after 4,158 search seconds, at about 22:46 on 5 October; the run wrapper’s native validation passed and the run stopped at 22:49, so the backtracking arm never ran. Its $E = 0$ state (6e569819; the certificate exported from it by the original engine is 6e923fbc) passed the same two checkers as the other three on the morning of 6 October.

The fifth diagram to be verified came from the cloud instance. Its lane f4 was a hold at

$T = 0.14$ and $\lambda = 1$ on a four-vCPU instance (c2d-highcpu-4, region us-east4-b), launched at about 21:23 on 5 October, seed 254004, from c6's own $E = 46$ state of 19:41 on 5 October (SHA-256 prefix 8e900bf9: one duplicated label of rank 11, with copies of degrees 3 and 5, and one missing label). At the 01:21 readout of the instance two of its three holds, f4 among them, stood at $E = 46$ and the third at $E = 92$, and its lane on the 19-curve recipe had not improved; f4's log then shows the same wandering as c21's between the two forms of $E = 46$, with excursions to $E = 92$ and once to $E = 138$, and it reached $E = 0$ at engine time 34,272 seconds, at 06:54 on 6 October, 9 h 31 min after launch, from the insertion form (two missing labels, of ranks 9 and 10, one bit apart, and no duplicate). Its zero-step reload in the original engine and the independent Python checker both passed at 07:35 on 6 October; the certificate exported from it is 5888e16a, and its canonical form (2670bfda) differs from those of the four diagrams above, so although it descends from lineage c6, as do three of the four, it is a fifth diagram, and the only one of the five found on a cloud instance rather than the laptop. In order of discovery the five diagrams are therefore the pilot diagram (about 22:46 on 5 October), c25 (00:00:09), c19 (02:15), c21 (04:58) and f4 (06:54 on 6 October); all five descend from the two twelve-day ramps, four from c6 (the pilot diagram from its $E = 184$ state, c25 from its $E = 92$ state, c21 from c20's $E = 46$ state and f4 from its own $E = 46$ state) and one, c19, from c9. When this version was prepared the other three laptop continuations stood at $E = 46$ and the two on the laptop running the 19-curve closure recipe had not improved on their starting energies.

The closing move. The pre-completion states, saved exactly as at 19, have two forms. For c25, c19 and f4 it is $E = 46$ with no duplicated region and two missing labels, of ranks 9 and 10, at Hamming distance 1 (for c19, after a cyclic relabelling; the two labels are 9 bits apart in the frame of the file): the form from which three of the 19-curve diagrams closed by a single lens insertion creating both missing labels at once. For c21 it is the opposite form, $E = 46$ with no missing label and two duplicated labels, both of rank 10, with copies of degrees 3 and 4 and of degrees 2 and 3, two bits apart in the frame of the file (no cyclic relabelling brings them closer): the form from which one 19-curve diagram (726d1fb0) closed by a single lens removal, though there the two duplicated labels were adjacent after a cyclic relabelling. So c21 closed by a deletion, removing both duplicated orbits at once, the first closure of that type at 23. The energies and label counts in the run logs are consistent with these moves; unlike at 19, the winning moves themselves were not recovered from the journal, so the mechanisms are inferred from the pre-completion states rather than recorded. The pilot diagram's run did not export a pre-completion state and did not record its closing move; its telemetry records the all-missing form ($E = 46$, two missing labels, no duplicate) at its first passage of $E = 46$ and again at its last periodic sample, 18 seconds before closure, so a state of that form preceded the closure, but whether the finishing move was an insertion is not established. A breadth-first search over count-neutral triangle flips from the campaign's $E = 46$ states, run while they were stalled, had found no finishing move to depth 6; the closures of c25, c19 and f4 came from the other side of that family, by insertion from states without duplicates, and c21's from a state of the family's own form, but one that the search had never been run on.

What did not work. In order, with the numbers of runs: a cloud fleet of 56 branches from the lowest laptop states (26–30 September) on which flat cooling at $T = 0.10$ and 0.08 froze at $E = 500$ – 600 , standard cooling sat at 230–276 for five hours, a melt at $\lambda = 0.35$ with the

degree-attacking proposal blew 16 branches up to $E \approx 22,000$, and the 138 floor held for 21 of the 56 branches with none lower in 35 hours; three laptop holds at $E = 138$ (c3 from 28 September and its restart c11 from 30 September, both stopped on 1 October, and c4, stopped on 3 October after about three days at 138); the eight-day ramps c5 and c8, with the 11 and 21 cloud continuations and 12 melting chains above; the 24 re-coolings from c5's end of ramp; two batches of four three-hour arms each of a memory-assisted (tabu and backtracking) variant of the engine written by Codex, in which memory off beat backtracking in both pairs of the first batch (best E 1,426 and 1,518 against 1,840 and 1,748) and no arm of the second went below $E = 92$ (the later paired pilot of the same engine from c6's $E = 184$ state is the run that found the pilot diagram, with its memory features inactive, so that run is evidence neither for nor against them); the exact flip search above (no finisher); six 600-second quenches at $T = 0.01$ from a stalled $E = 46$ state, all of which stayed at 46; three continuations on the 19-curve closure recipe; two gated fresh starts stopped at $E \approx 780$ and 550; and 112 fresh starts on two 56-vCPU spot instances from 29 September, on a four-day ramp, one-day hold, one-day cooling and a hold at $T = 0.14$, half from each scaffold, none of which had gone below $E = 138$ when this version was prepared.

Verification and distinctness. Each of the five states was checked as at 19: a zero-step reload in the original full-state engine with its structure check (missing 0, duplicates 0, energy 0), then the independent Python checker on the exported faces ($V = 8,388,608$, $E = 16,777,212$, $F = 8,388,606$, Euler characteristic 2, every edge in two faces, every face a transversal cube square, one rotation cycle per vertex, both sides of every curve connected, invariant under label rotation), and then the public checker of the data repository on the certificate JSON (all five PASS, in 693, 734, 610, 633 and 614 seconds). All five are non-monotone (Section 6). c25 and c19 share 827,149 of their 8,388,606 faces, about a tenth, the same proportion as between the fresh-start diagrams at 19 (the face-set overlaps of the other three certificates were not computed). Distinctness was established in two ways. First, by the method used for the 17- and 19-curve certificates: the canonical form of each certificate, the minimum over all $4n = 92$ maps (cyclic relabelling, mirror image, exchange of the poles) of the hash of its sorted relabelled face list, was computed for c25, c19, c21, the pilot diagram and f4 (400e5279, 337076ae, 56f659b4, 029acb28 and 2670bfda), and the five forms differ pairwise: five files, five isomorphism classes. Second, the region-degree histograms of the five certificates, computed independently by Codex from the certificate files, differ for every one of the ten pairs (for example 2,951,107, 2,950,003, 2,937,192, 2,947,197 and 2,944,046 regions of degree 3 in c25, c19, c21, the pilot diagram and f4); the degree histogram is unchanged by any relabelling of the curves, not only the 92 maps, so the five diagrams are pairwise non-isomorphic under arbitrary relabelling (Table 1), although c25, c21, the pilot diagram and f4 all descend from lineage c6.

6 Properties

All eighteen diagrams are non-monotone in the sense of Bultena, Grünbaum and Ruskey [6]: in the first 19-curve certificate, 3,590 of the 27,594 region orbits (13%) lack a neighbouring region of rank one lower or one higher, the 17-curve certificates behave the same way, and in the 23-curve certificates c25, c19, c21, 6e923fbc and f4 52,742, 52,279, 52,597, 52,518 and 52,501 of the 364,722 region orbits (14%) lack such a neighbour (the two poles, fixed by the rotation, are monotone and are not counted among the orbits here or in the 19-curve figure). Since

monotone diagrams are exactly those that can be drawn with convex curves [6], none of them has a convex drawing. Every previously published simple symmetric Venn diagram with $n \geq 11$ is monotone, and the crossing-sequence representation used to find them [15, 14] enumerates monotone diagrams by construction, so the diagrams reported here were outside the search space of those methods; Mamakani and Ruskey themselves noted that the size of their search sequences grows about tenfold from 13 to 17 curves [15]. Minimum region degree is 3 (a degree-2 region is impossible in a simple Venn diagram with $n \geq 3$); in the first 19-curve certificate the degree histogram is 187,340 regions of degree 3, 199,120 of degree 4, 98,686 of degree 5, and a tail to degree 16, with the two poles of degree 19 and every rank-1 and rank-18 region of degree 11.

7 Data

Certificates, checker transcripts, the search code with its checksums, the independent checkers and both Lean developments are at <https://github.com/dzoba/venn17> and archived on Zenodo under the concept DOI <https://doi.org/10.5281/zenodo.22885649>, which resolves to the latest version. The five 23-curve certificates are 889 MB each and are not in the git repository: they are carried as gzip-compressed assets (about 103 MB each) of the repository’s release v1.3 and in the Zenodo record of version 1.3, <https://doi.org/10.5281/zenodo.23189412>, with the SHA-256 sums of both the `.json` and the `.json.gz` files in the repository’s README and of the `.json` files in `certificates/SHA256SUMS`. For all three sizes the published certificates, the checker transcripts and the checksums are the public record; the pre-completion states (of the 17-, 19- and 23-curve diagrams alike), the run logs and the dated run records of the 23-curve campaign (the launch manifest and the cloud log) are kept by the author.

Tool and computational resource disclosure

This work was carried out by two AI systems working under the author’s direction, over five days for the 17- and 19-curve diagrams and a further sixteen days, from 20 September to 6 October 2026, for the 23-curve diagrams, and this note discloses their roles in full, in the spirit of the Leiden Declaration on Artificial Intelligence and Mathematics. Claude (Anthropic) formulated the dual-map encoding, designed and implemented the search (the relaxed walk, its moves, energy and annealing schedules), diagnosed the failure at 19 from the run records, designed and implemented the added proposal class and the capture of pre-completion states, ran the searches and the verification, performed the analyses reported here and drafted this note; at 23 it also ran the campaign day to day (the laptop lineages and the cloud fleets, their schedules, the defect censuses that identified the crust obstruction, and the choice of the states to continue from), keeping a dated log of every launch and stop. Codex (OpenAI) constructed the resolved Griggs–Killian–Savage starting states and the orbit-safe 23-curve scaffold, wrote the quotient-native rewrite of the search engine used at 23, wrote independent validators for the 17- and 19-curve certificates, identified the need for the bigon moves, built exact local solvers used to diagnose the plateaus at 19 and at 23, built the memory-assisted engine variants tested at 23 and ran the pilot of that engine whose ordinary arm found the first of the 23-curve diagrams in order of discovery, and ported the Lean development to 19 curves. The author set the goals and strategy, decided what to pursue and what to abandon, approved the launches and stops recorded in the campaign log, supplied the computing resources, and is responsible for the content. Computing:

one 16-core laptop (Apple M4 Max, 128 GB) throughout, on which four of the five 23-curve diagrams were found, the fifth (f4) on a four-vCPU cloud instance; for the 19-curve runs cloud instances of up to 24 vCPUs each, never more than about a hundred vCPUs at once and none for longer than two days, at a total cloud cost under five hundred dollars; for the 23-curve campaign spot and standard instances of up to 64 vCPUs each, at most four at once (about 170 vCPUs), the longest-lived running from 29 September to the time of writing. Software: the search engine, checkers and plotter in the data repository (C++ and Python with NumPy); Lean 4 with Mathlib for the formal proofs; `tectonic` for this document.

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n	certificate	SHA-256 prefix	non-isomorphic
17	venn17-local-c3-s2	c178d7bd	yes
17	venn17-gcp-s12	87f981b8	yes
17	venn17-gcp-s14	eec8cd55	yes
17	venn17-gcp-s16	431857aa	yes
19	venn19-closure-s196002	fc7ad34	yes
19	venn19-closure-s196004	285de8ea	yes
19	venn19-closure-s195001	a09ceae0	yes
19	venn19-closure-s196001	56e7e1a1	yes
19	venn19-closure-s196007	312e8c88	yes
19	venn19-closure-s195002	726d1fb0	yes
19	venn19-fresh-s192015	45a8aee9	yes
19	venn19-fresh-s190002	b46bf5d0	yes
19	venn19-fresh-s192007	d5b20c27	yes
23	venn23-c25-s230025	adc5a02e	yes
23	venn23-c19-s230019	5dfddac2	yes
23	venn23-c21-s230021	e6849a0b	yes
23	venn23-c6p-s1065101	6e923fbc	yes
23	venn23-f4-s254004	5888e16a	yes

Table 1: The eighteen certificates. Within each size the diagrams are pairwise non-isomorphic under cyclic relabelling of the curves, mirror image and exchange of the two poles, by canonical forms compared over all $4n$ such maps; the five 23-curve certificates are moreover pairwise non-isomorphic under arbitrary relabelling of the curves, by their region-degree histograms, which differ for every pair. As face sets the nine 19-curve certificates differ pairwise by between 25,460 and 947,188 faces of 524,286; each of the three fresh-start certificates shares only about a tenth of its faces with any other. The 23-curve certificates `adc5a02e` and `5dfddac2` share 827,149 of their 8,388,606 faces (9.9%) in the frame in which they were found, and the same under every cyclic relabelling, since each face set is invariant under label rotation; the canonical forms of `adc5a02e`, `5dfddac2`, `e6849a0b`, `6e923fbc` and `5888e16a` over all $4n = 92$ maps (`400e5279`, `337076ae`, `56f659b4`, `029acb28` and `2670bfda`) differ pairwise. The prefixes of the six 19-curve closure rows are those of the raw search states from which the certificates were exported (the JSON form of `fc7ad34` is `ed26b3ba`); all other prefixes, including those of the three fresh-start 19-curve rows and of the 23-curve rows, are those of the certificate JSON files, the form in which they are published, and the JSON prefixes of all eighteen are listed in `certificates/SHA256SUMS`.